MILITARY SPECIFICATION

SPARK PLUG: SHIELDED, FINE WIRE IRIDIUM ELECTRODE, TYPE 1, 2, 3, 4, FOR AIRCRAFT RECIPROCATING ENGINES

This specification is approved for use by the US Army Aviation Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers type 1, 2, 3, 4, fine wire iridium electrode spark plugs for:

AIRCRAFT	ENGINE	SPARK PLUG TYPE
T-41B	IO-360-D	2, 4
TH-55A	HIO-360-B1A	2, 4
U-10A	GO-480-G1D6	2, 3, 4
U-8F	IGSO-480-A1A6 (O-480-3/3A)	1, 2, 3, 4
U-3A	O-470-M	1, 3
U-3B	IO-470-D	1, 3
T-42A	IO-470-L	1, 3

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Aviation Research and Development Command, ATTN: DRDAV-ERS, St. Louis, Missouri 63120, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-L-6082	Lubricating Oil, Aircraft Reciprocating (Piston) Engine
MIL-S-7886	Spark Plug, Shielded, Aircraft Reciprocating Engine, General Specification for

STANDARDS

MILITARY

MIL-STD-414	Sampling Procedure and Table for Inspection by Variables for Percent Defective
MS17840	Gasket - Spark Plug

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CID A-A-711 Dry Cleaning Solvent

(Copies of the commercial item description required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM D 484 Solvent, Dry Cleaning Hydrocarbon

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Rece Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the test of this specification and the references cited herein, the test of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Qualification. The spark plug shall be a product which has passed the qualification tests in accordance with MIL-S-7886 for the engine-airframe combination specified in Table I and is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4. and 6.).

3.2 Performance Reliability. Spark plug shall be capable of operating satisfactorily without fouling or other evidence of electrical or mechanical failure when subjected to the flight tests specified in (4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptance Inspections. Acceptance Inspections shall be in accordance with MIL-S-7886. The Sampling Plan for Group B inspection shall be in accordance with MIL-STD-414, Method B, C. or D; AQL-1.0% for the Lower Limit and AQL-2.5% for the Upper Limit, Level 1. Lot size not to exceed one continuous production run of a specific design for a specific contract.

4.2 Test conditions.

4.2.1 Engine. Only an engine in proper mechanical and electrical condition, having sufficient time before the next overhaul for the flight test involved, shall be used for the testing of the spark plugs. Bushing threads in the cylinder shall be cleaned of carbon and other foreign matter. The maintenance inspection of the engine shall also include a check on (1) the compression and (2) the ignition components including coils, breaker points, condensor, ignition harness, distributor, and magneto and position.

4.2.2 Identification of Plugs. Prior to installation of each complete set of test plugs of the same part number, each plug shall be permanently marked by etching to indicate cylinder number and position.

4.2.3 Gasket. Whenever a plug is installed, a new gasket conforming to MS17840 shall be used.

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4.2.4 Torque. Plugs shall be tightened to the torque specified for the test condition preferably to the high limits, as specified in MIL-S-7886.

4.2.5 Lubrication. Whenever a plug is removed for inspection, the threads shall be lubricated with engine oil conforming to grade 1100 of MIL-L-6082, prior to its reinstallation.

4.3 Tests. Spark plugs shall be tested under controlled, and under other than controlled flight conditions, as specified below. Engine-airframe combinations, fuel, electrode gap, flight hours, and allowable number of spark plug failures for each engine, shall be as specified in Table I.

TABLE I - Engine-Airframe Combinations

ENGINE	AIRCRAFT	FUEL	ELECTRODE GAP (IN.)	FLIGHT HOURS	PLUG FAILURES PER ENGINE
IO-360-D	T-41B	100LL	.015/.018	600	2
HIO-360-B1A	TH-55A	100LL	.015/.018	600	2
GO-480-G1D6	U-10A	100LL	.015/.018	600	2
IGSO-480-A1A6 (O-480-3/3A)	U-8F	100LL	.015/.018	600	2
O-470-M	U-3A	100LL	.015/.018	600	2
IO-470-D	U-3B	100LL	.015/.018	600	2
IO-470-L	T-42A	100LL	.015/.018	600	2

4.3.1 Controlled Flight. At the beginning of the controlled flight tests, and at approximately every 100 hours of flight time thereafter, the following three tests, for both rotary wing and fixed wing, shall be successfully accomplished three times:

4.3.1.1 Rotary Wing.

4.3.1.1.1 Lifting and Hovering Capability. Engine shall be allowed to idle in auto rich for 15 minutes, using normal defouling procedures. Engine shall then be capable of lifting and hovering the craft with its maximum military load until highest permissible engine cylinder head temperature is reached.

4.3.1.1.2 Maximum Rate of Climb and Power. After engine cools to normal operating temperature, it shall be capable of taking off and maintaining maximum rate of climb to the aircraft maximum operating altitude, followed by a 15 minute cruise at low power to cool and stabilize engine temperatures. Then a level course maintained at maximum except takeoff (METO) for 5 minutes, followed by maximum endurance power for 30 minutes.

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4.3.1.1.3 Cruise Power and Autorotation. Engine shall be cruised at low power for 15 minutes, to cool the engine, and autorotated to safe recovery altitude, followed by level flight at cruise power for 30 minutes.

4.3.1.2 Fixed Wing.

4.3.1.2.1 Maximum Rate of Climb with Maximum Military Load. Engine shall be allowed to idle in auto rich for 15 minutes, using normal defouling procedures. Engine shall then be capable of maintaining maximum rate of climb with maximum military load from point of takeoff to the aircraft maximum operating altitude.

4.3.1.2.2 Cruise and Maximum Endurance Power. The engine shall then be operated for 15 minutes at approximately 65 percent power in cruise to cool and stabilize engine temperatures, followed by maximum except takeoff (METO) power for 5 minutes, and then at maximum endurance power for 30 minutes.

4.3.1.2.3 Cruise, Glide, and Level Flight. The cruise shall be at approximately 65 percent power for another 15 minutes to cool the engine, followed by a two minute glide at 20 percent power, and again at approximately 65 percent power in level flight for 15 minutes.

4.3.2 Other Than Controlled Flight. Flight time other than controlled shall be under normal service conditions and in accordance with applicable technical data relating to performance of typical maneuvers normally associated with observations, resupply, communication relay, and similar type missions.

4.4 Inspections.

4.4.1 Intervals. Plugs shall be tested without their removal by using an ignition analyzer at 50, 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, and 600 hours, to determine whether the plugs in each cylinder function properly.

4.4.1.1 Malfunctioning Plugs. Any plugs showing indication of malfunctioning shall be removed from the engine and visually inspected. If oil-fouled, the plugs shall be cleaned by soaking the firing end cavity for 30 minutes in solvent conforming to CID A-A-711 and ASTM D 484, then draining and thoroughly drying them with a filtered airblast. A light abrasive blast cleaning is permissible if necessary to remove hardened oil deposits. In addition, the ceramic shielding barrel shall be thoroughly cleaned. The elctrode gaps shall be placed in a spark plug tester such as the AV 18-1 manufactured by AC Spark Plug Div. of General Motors Corp., or equivalent. Tester shall be connected to a single phase 110-120 volt, 60 cycle power supply outlet. Nitrogen gas shall be used in the pressure chamber: for spark plug electrode gaps under .020 inch, 200 psi; for gaps over .020 inch, 175 psi. If plug fires under these conditions, it shall be reinstalled in same position in the engine, using a new gasket conforming to MS17840.

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4.4.1.2 Plug Failures. Plugs other than oil-fouled, or showing obvious mechanical failure, shall be tested in a spark plug tester (see 4.4.1.1). Any plug failing to fire shall be considered a failed plug. Plug failures due to other than normal flight test conditions are not to be considered, plug failures shall be replaced with like new items. Cracked ceramic insulators, electrode failures, chronic fuel and lead fouling, or other bonafide failures are considered plug failures.

4.4.2 Final Examination. At the completion of 600 flight hours, plugs shall be removed from the engine and given a final inspection. This shall include visual inspection, final gap measurement, and testing in spark plug tester as specified in 4.4.1.1. Results of these inspections shall be shown on test report indicating the performance of each set of spark plugs under tests and examinations specified herein.

4.5 Test Stoppage. If more than the allowable number of plug failures occur before completion of flight hours scheduled (See Table I) test shall be discontinued.

5. PACKAGING

5.1 Preparation for Delivery. Preparation for delivery shall be in accordance with Section 5 of MIL-S-7886.

6. NOTES

6.1 Intended Use. The spark plug types intended for use with certain engine-airframe combinations are specified in paragraph 4.3.

6.2 Ordering Data. Ordering data shall be in accordance with MIL-S-7886.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Headquarters, US Army Aviation Research and Development Command, ATTN: DRDAV-ERS (Technical Services Division, Standardization Branch), 4300 Goodfellow Boulevard, St. Louis, Missouri 63120. Information pertaining to qualification of products may be obtained from DRDAV-D (Directorate for Development and Qualification) of the same above address.

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